ICS 4111: Embedded Systems & IoT

Practical Exercise 3: Embedded System Design

Instructions:

In your assigned groups for the semester, you have been provided with the following

Components:

- A microcontroller
- A LED
- A breadboard
- One or two resistors
- A multimeter
- A potentiometer

Objective: Practical demonstration of Ohm's Law. For this exercise, you are expected to document your findings by responding to the reflection questions, taking images of your setups for each experiment. These can be presented as PDF document (s).

Group Members & Images

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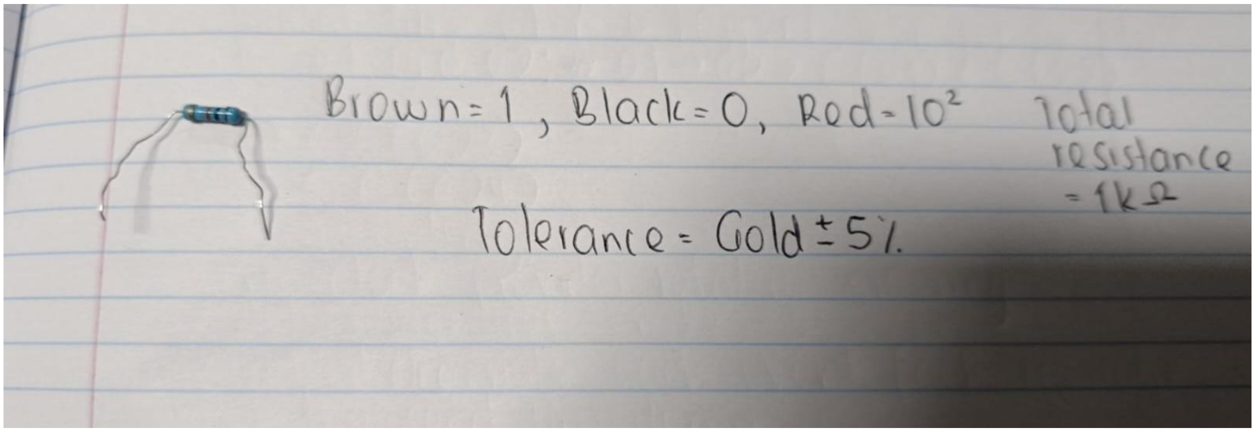
Experiment 1:

1. Pick one resistor, take a picture of it and determine its resistance value based on the colour code. Write down the resistance value.



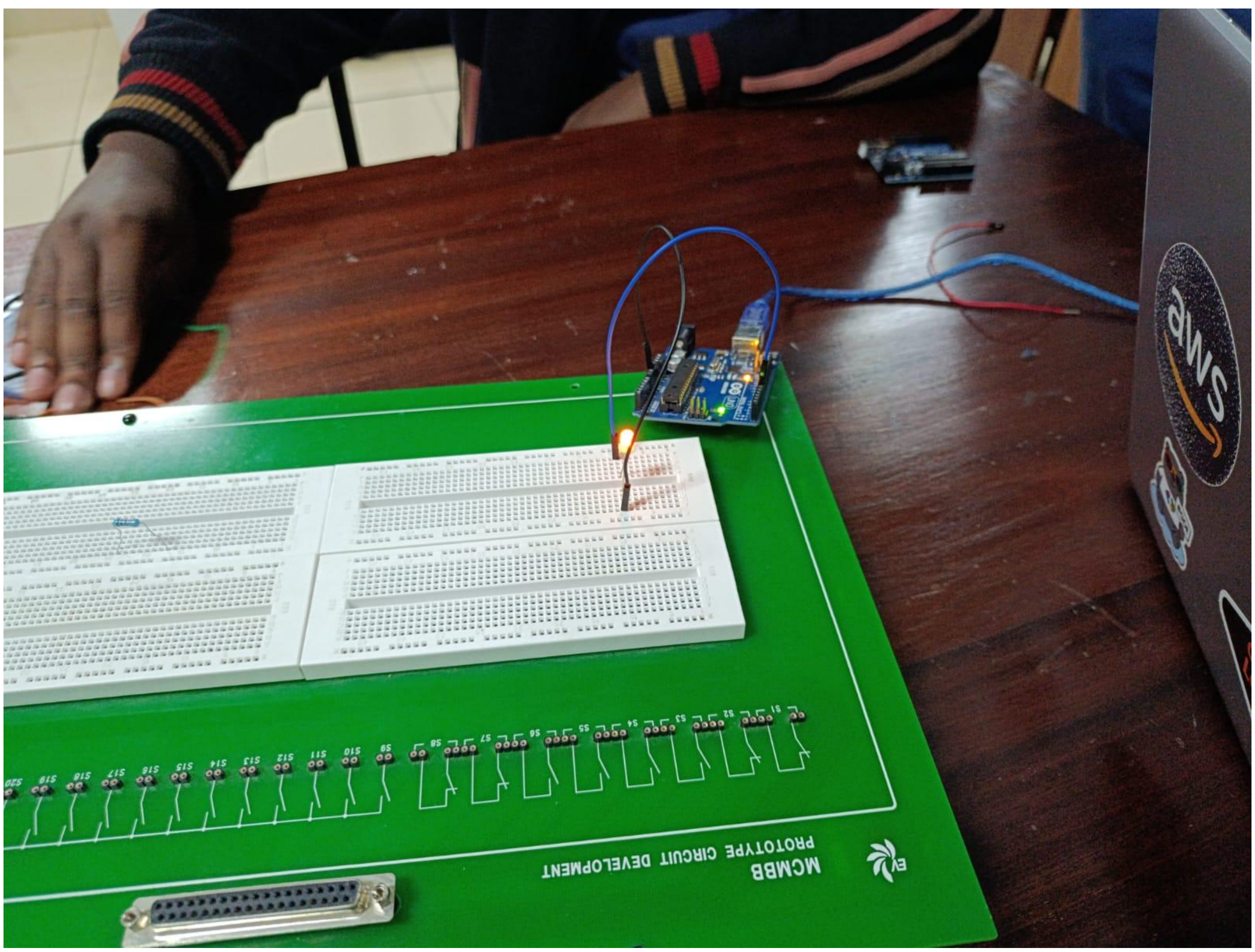
2. Using a multimeter, measure the resistance value emitted by the resistor. Take a picture of this setup and write down the measured resistance.

Experiment Reflection: Where does the difference in the resistance values arise from?

The multimeter has an error margin +-(0.5% + digits)

Experiment 2:

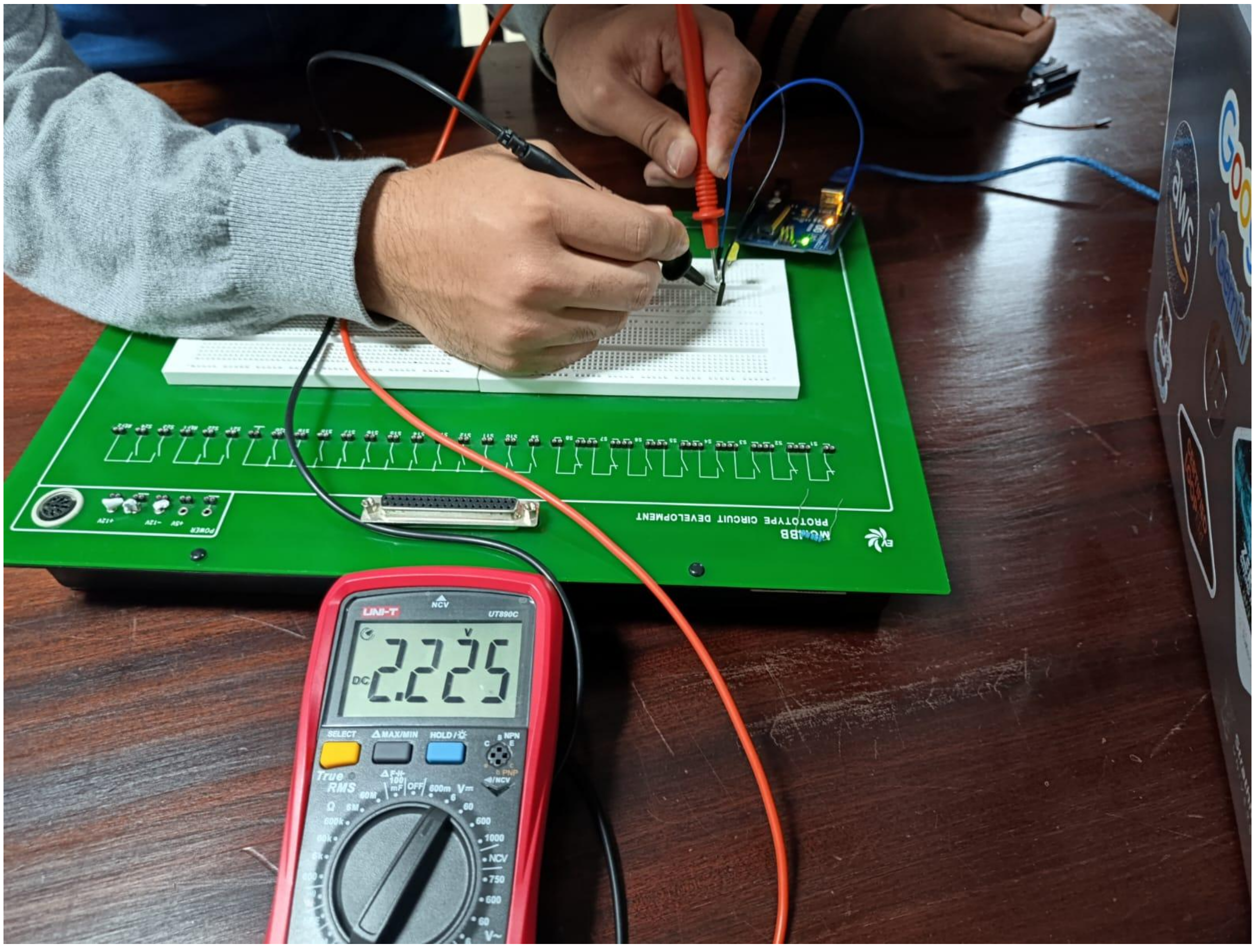
1. Using the breadboard, microcontroller, and LED, recreate the following schematic. Take a picture of the physical implementation.



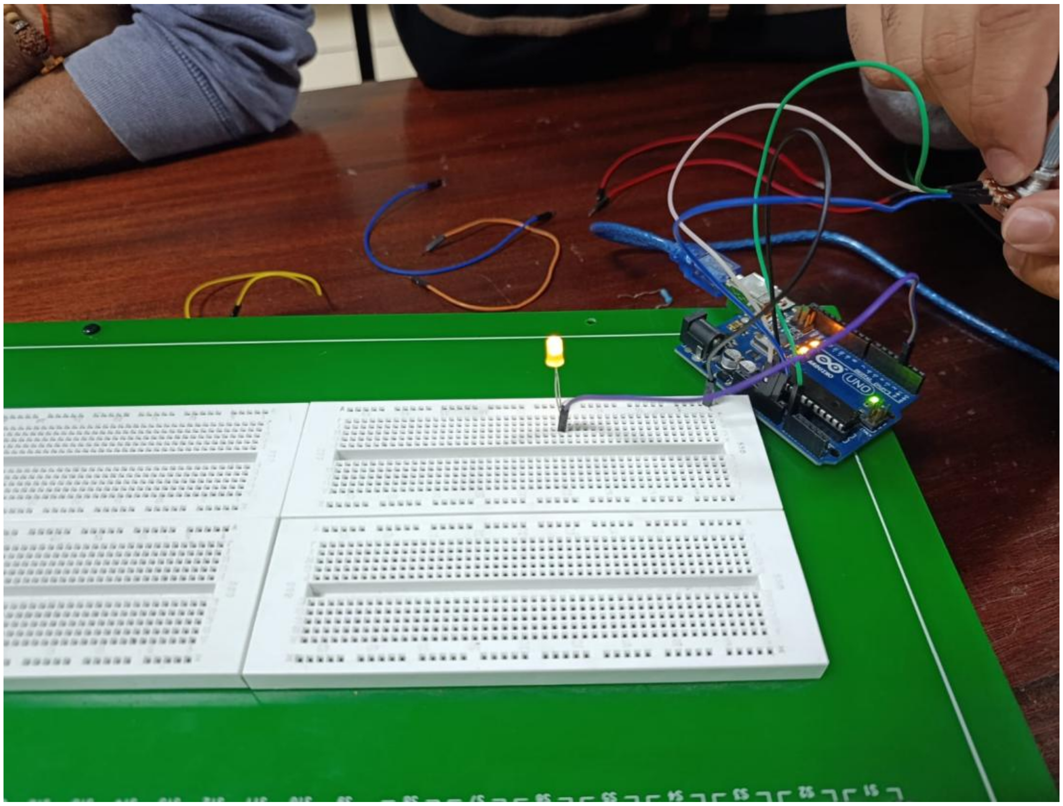
2. In Arduino IDE, deploy the Blink program which is usually inbuilt in Arduino and upload to your board. Congratulations you just created your first sketch! :)

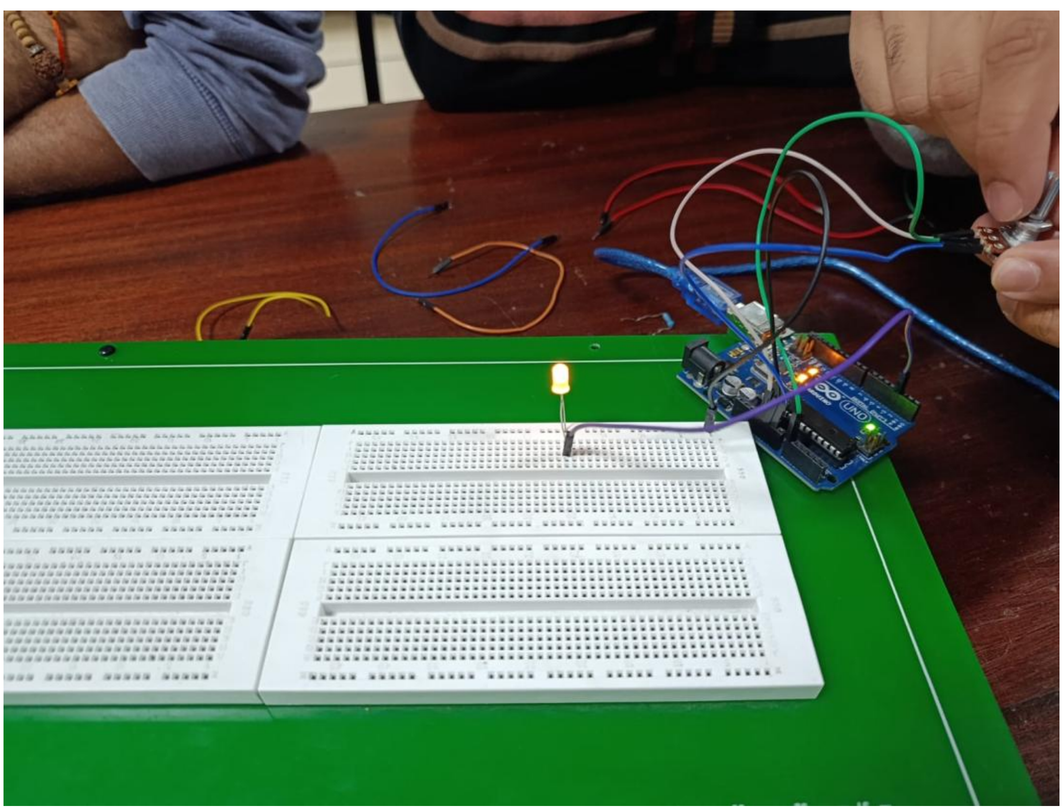
```
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(6, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(6, HIGH);  // change state of the LED by setting the pin to the HIGH voltage level
  delay(1000);                      // wait for a second
  digitalWrite(6, LOW);   // change state of the LED by setting the pin to the LOW voltage level
  delay(1000);                      // wait for a second
}
```

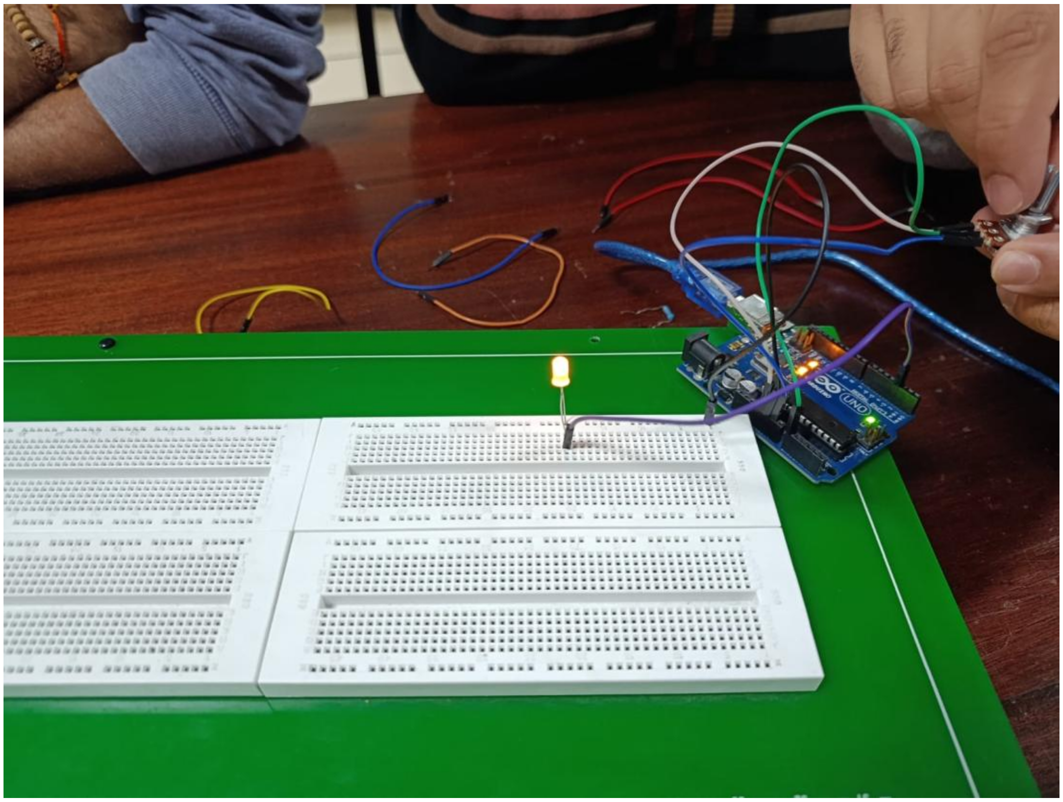
3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered.



4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate Pulse Width Modulation (PWM) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness. You can refer to these or any other suitable resources to guide on the connection: Using PWM to Control the Light Intensity of a LED - Instructables; Arduino LED Dimmer (Potentiometer + PWM)

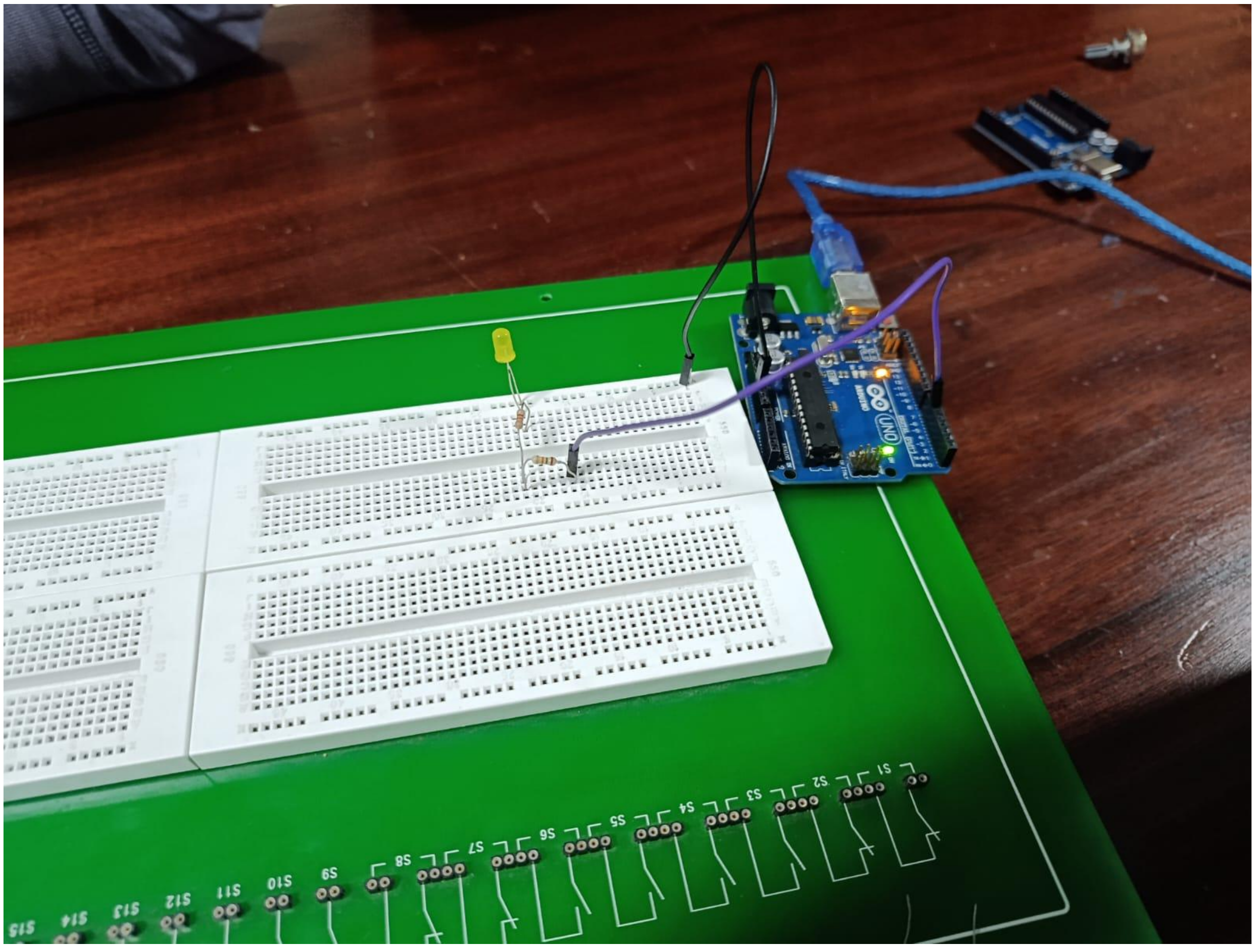






Experiment 3:

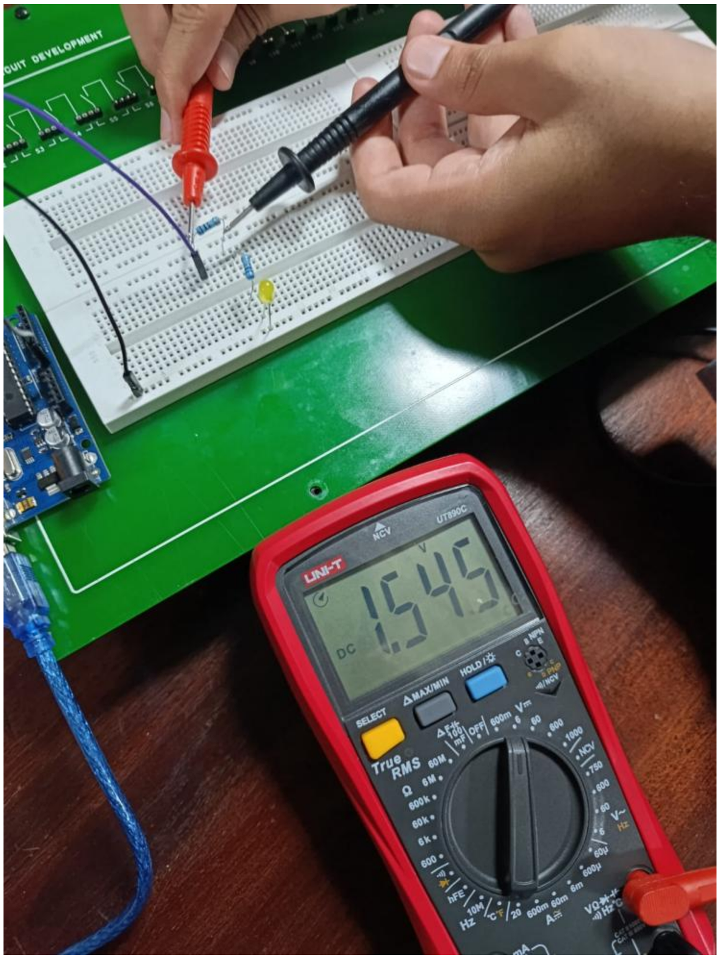
1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical implementation.

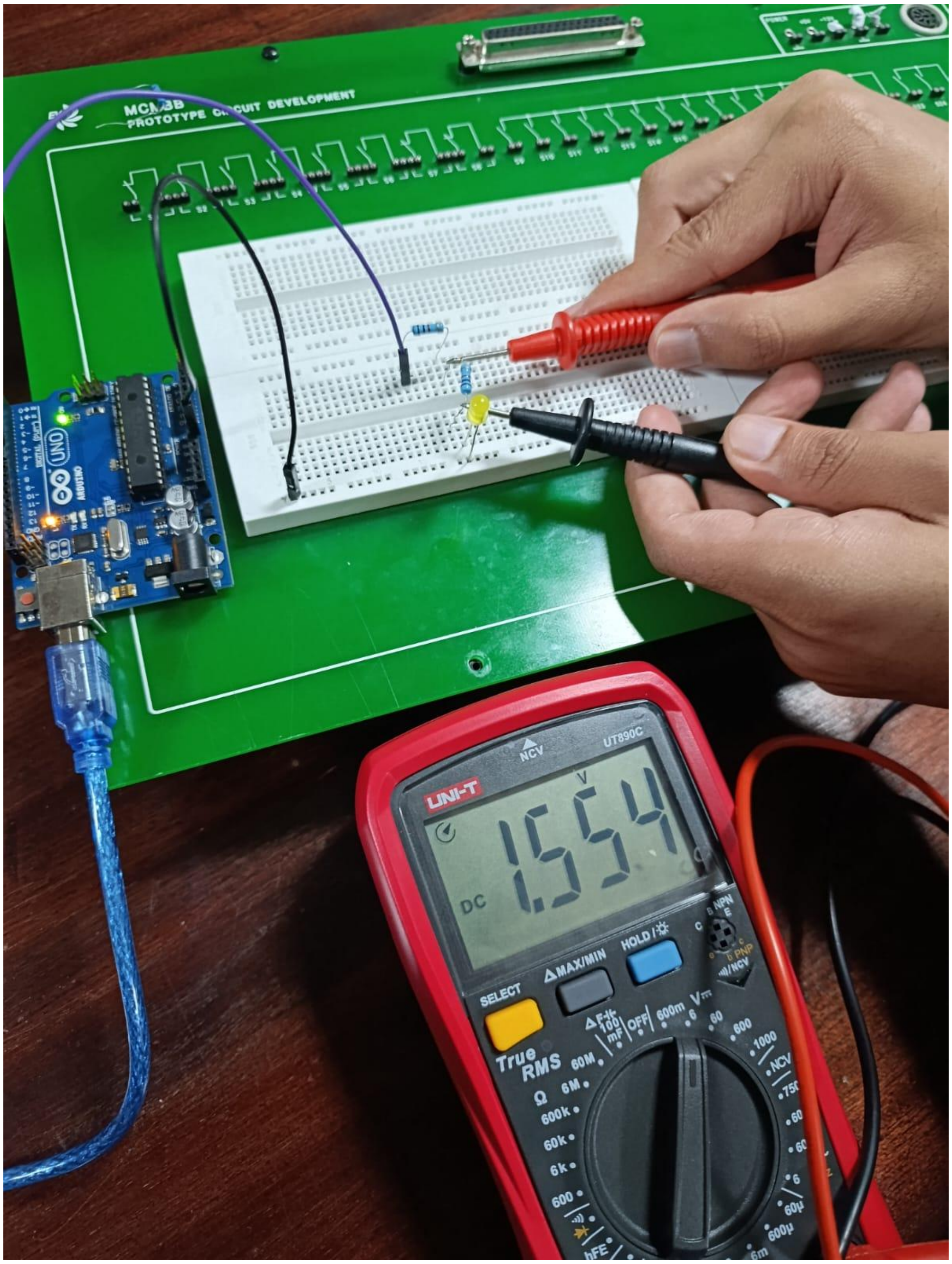


2. In Arduino IDE, re-deploy the Blink program and upload to your board.

```
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(6, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(6, HIGH);  // change state of the LED by setting the pin to the HIGH voltage level
  delay(1000);                      // wait for a second
  digitalWrite(6, LOW);   // change state of the LED by setting the pin to the LOW voltage level
  delay(1000);                      // wait for a second
}
```

3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.

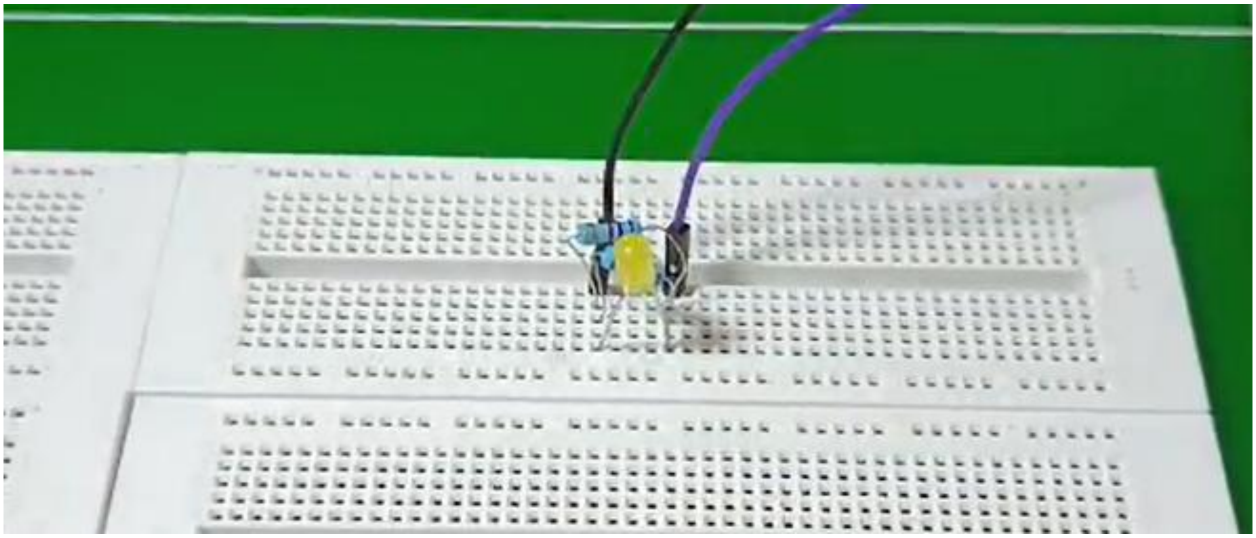




Experiment Reflection: Use Ohm’s Law to estimate the current passed at each resistor.

Experiment 4:

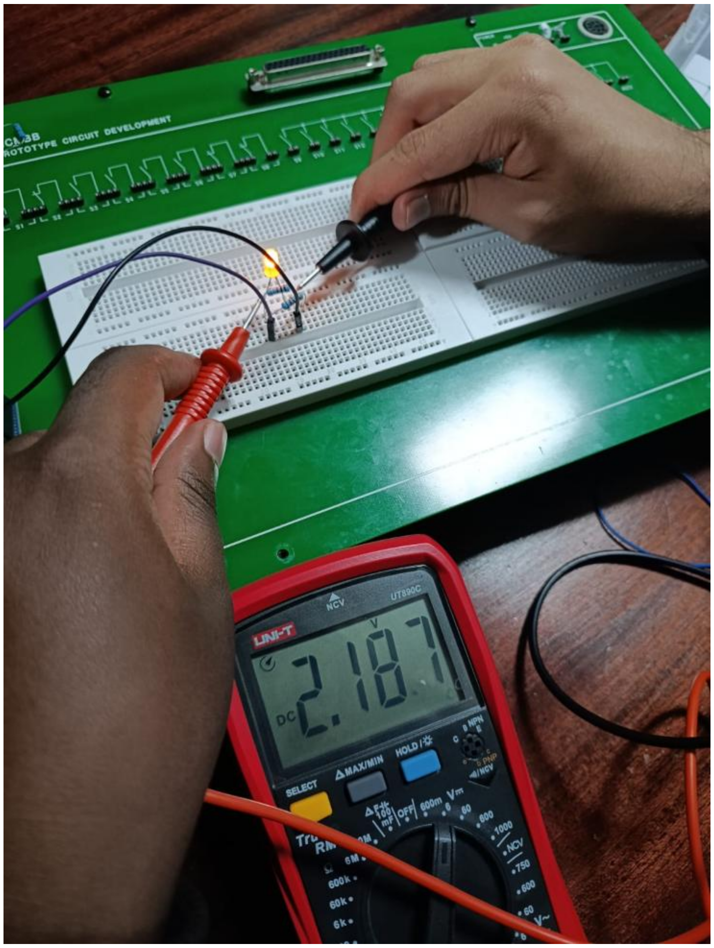
1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the Blink program and upload to your board.

```
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(6, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(6, HIGH);  // change state of the LED by setting the pin to the HIGH voltage level
  delay(1000);                      // wait for a second
  digitalWrite(6, LOW);   // change state of the LED by setting the pin to the LOW voltage level
  delay(1000);                      // wait for a second
}
```

3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.


PROTOTYPE CIRCUIT DEVELOPMENT
UNI-T
UT890C
NCV
2.187
DC
SELECT
MAX/MIN
HOLD